MIL-C-24231/29(SH)
15 May 1987
SUPERSEDING
NAVSHIPS DRAWING
815-1197254(IN PART)
26 January 1977

MILITARY SPECIFICATION SHEET

CONNECTOR, PREFORM MOLDED INSERTS, FOR MOLDED PLUGS, TYPE I

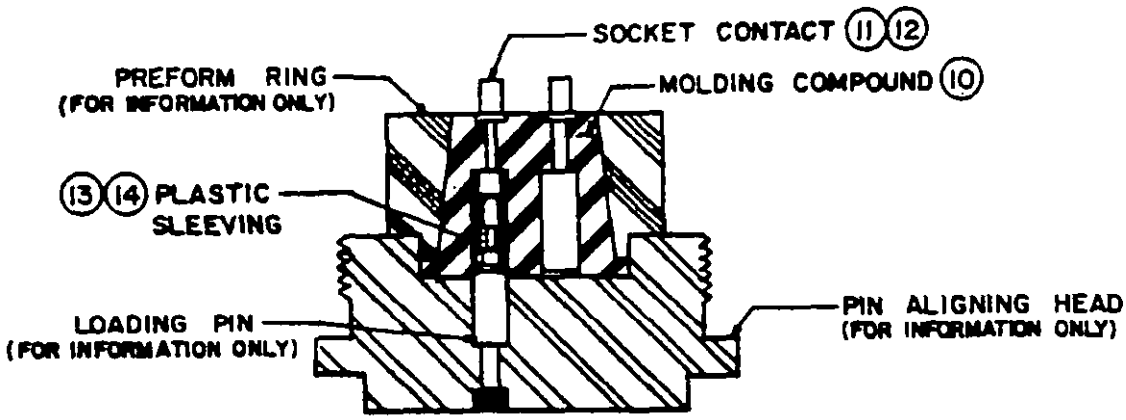
This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the inserts described herein shall consist of this specification and the latest issue of MIL-C-24231.

AMSC N/A FSC 5935

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SH 1318[7]

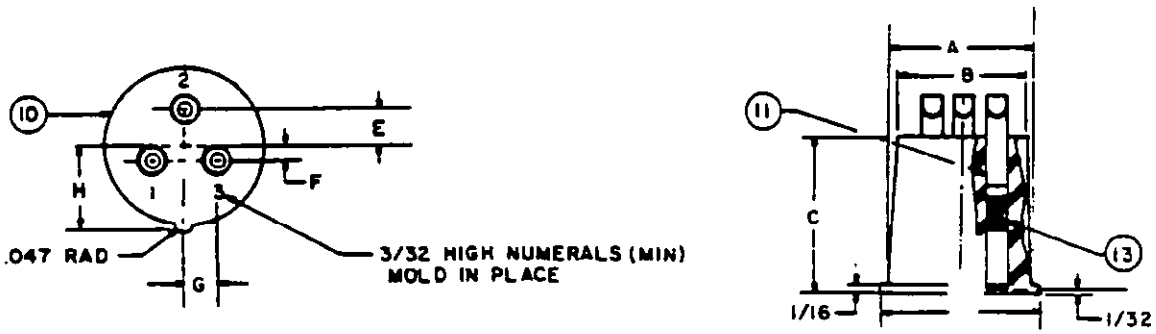
TABLE I. Insert assembly.

Item no.	Name	Use with plug spec MIL-M-24231/	Symbol no.
01	Insert assembly	M24231/1-001 M24231/2-001	713 713.5
02		M24231/1-002 M24231/2-002	713.6 713.8
03		M24231/23-001	713.9
04		M24231/3-001	713.1
05		M24231/3-002	713.7
06		M24231/4-001 M24231/4-006	713.2
07		M24231/4-002 M24231/4-007	713.3
08		M24231/4-003 M24231/4-008	713.4
09		M24231/4-004 or 005 M24231/4-009	713.10 --- ---
15		M24231/1-003 M24231/2-003	--- ---

FIGURE 1. Preform insert assembly, typical.

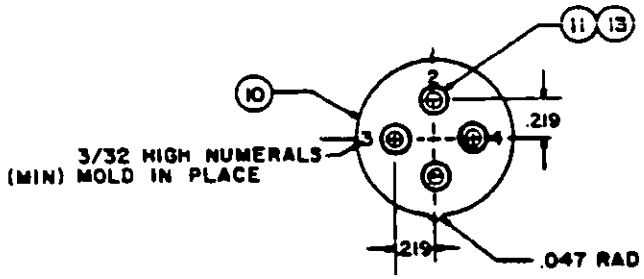
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ITEM NO	DIA A	DIA B	DIM C	DIA D	DIM E	DIM F	DIM G	DIM H
01	0.750	0.625	0.813	0.875	0.156	0.077	0.135	0.484
02	1.063	0.875	1.188	1.188	0.270	0.135	0.233	0.641
03	1.063	0.875	1.188	1.188	0.270	0.135	0.233	0.641



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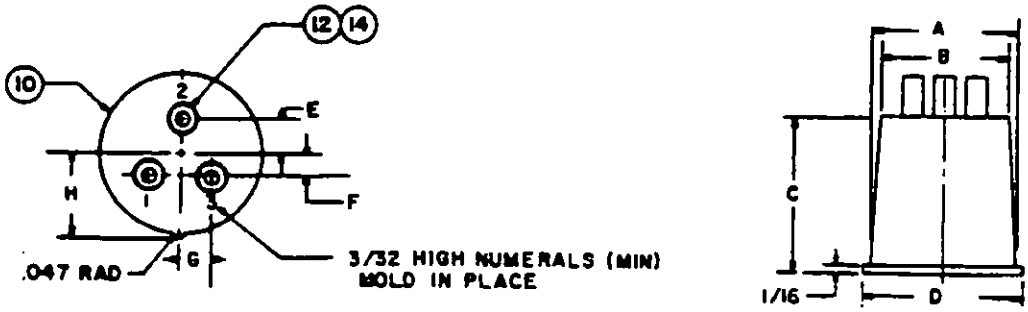
FIGURE 2. Insert assembly, (3-pin). (01)



SH 131899

FIGURE 3. Insert assembly, (4-pin). (02)

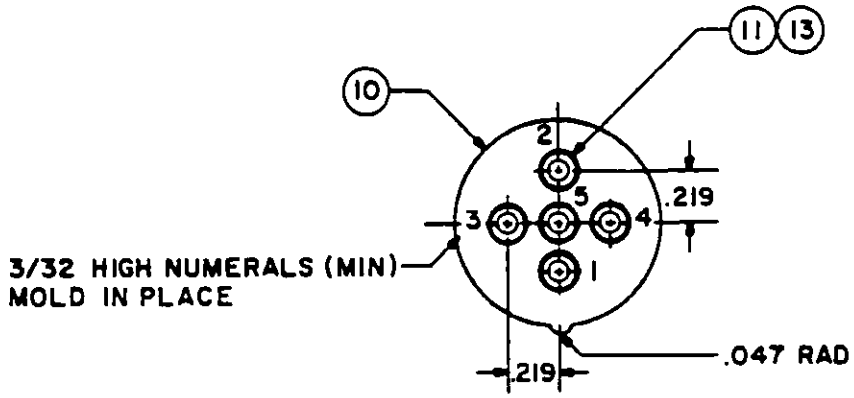
(same as item 01, except as shown).



SH 131900

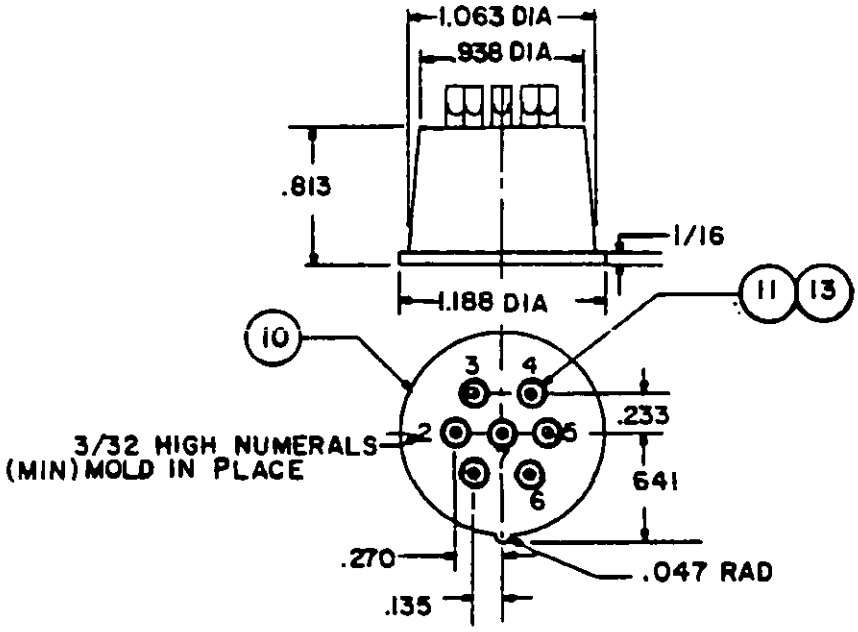
FIGURE 4. Insert assembly, (3-pin, no. 12). (03)

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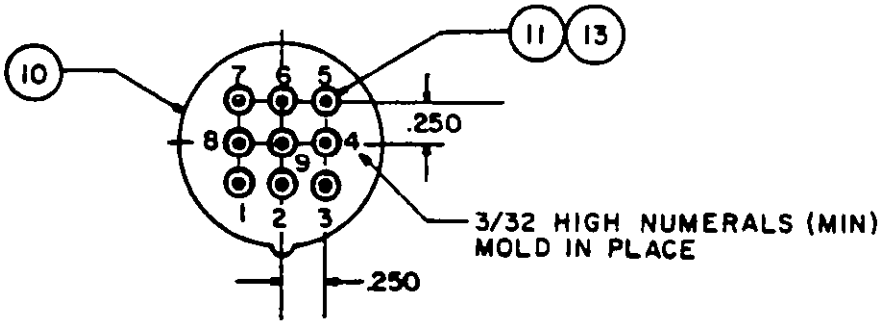
SH 131901

FIGURE 5. Insert assembly (5-pin). (15)
(same as item 02, except as shown).



SH 131902

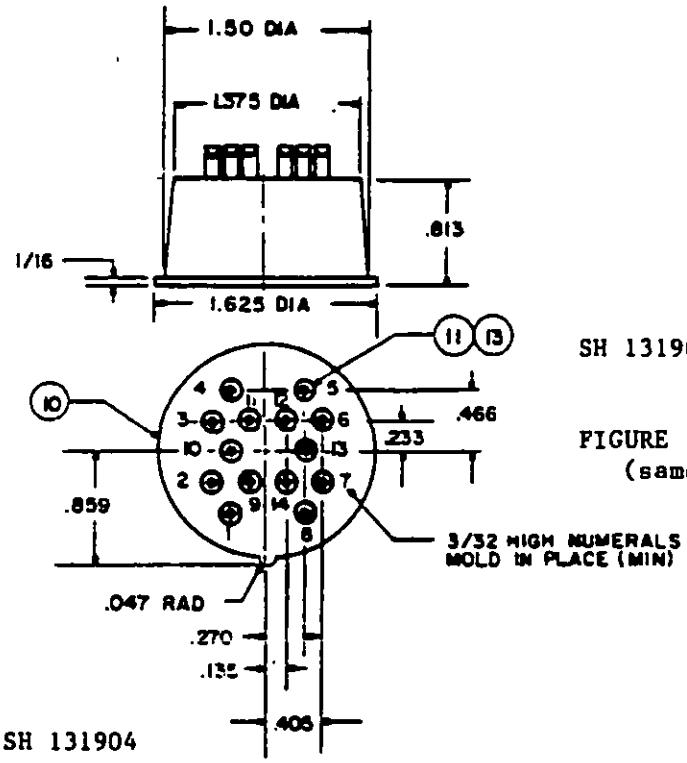
FIGURE 6. Insert assembly, (7-pin). (04)



SH 131903

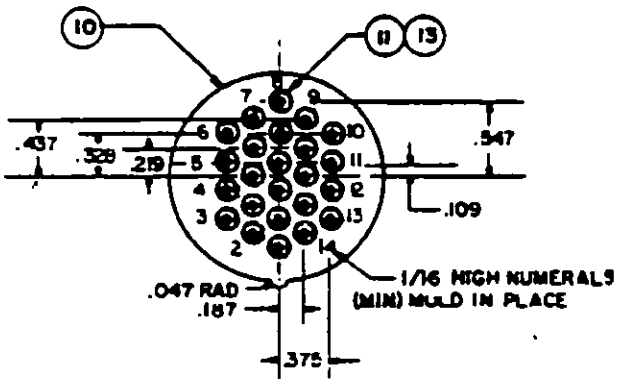
FIGURE 7. Insert assembly (9-pin) (05)
(same as item 04, except as shown)

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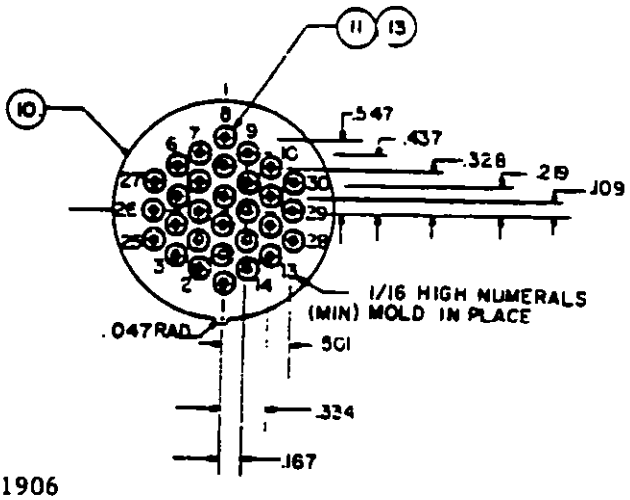
SH 131904

FIGURE 8. Insert assembly, (14-pin). (06)



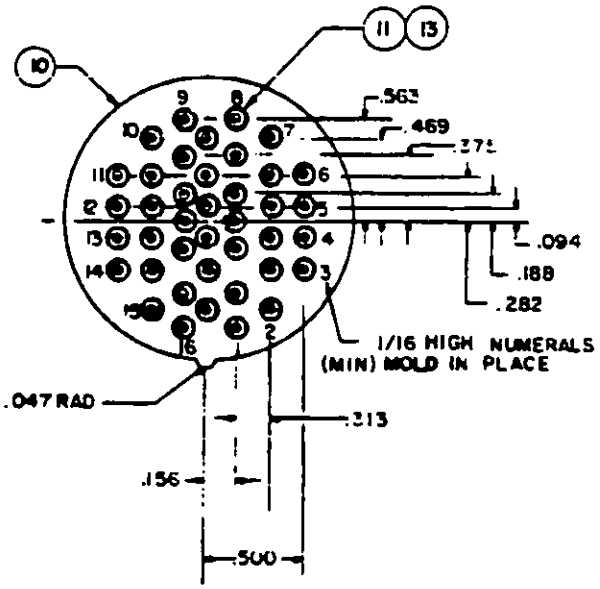
SH 131905

FIGURE 9. Insert assembly, (24-pin). (07)
(same as item 06, except as shown).



SH 131906

FIGURE 10. Insert assembly, (30-pin). (08)
(same as item 06, except as shown).



SH 131907

FIGURE 11. Insert assembly, (40-pin). (09)
(same as item 06, except as shown)

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REQUIREMENTS:

1. The molded inserts shall be used when molding symbol nos. 713, 713.1, 713.2, 713.3, 713.4, 713.5, 713.6, 713.7, 713.8, 713.9 and 713.10 plugs - plus the 5-conductor plugs conforming to the applicable specification sheet of MIL-C-24231 (see table I).
2. Design. Dimensions and configuration: See figures 1 through 11. Locating dimension for socket contacts for reference only, since the location will be controlled by the applicable pin aligning head.
3. Materials: As specified in table III.
4. General information. Assignment of military part numbers and general information shall be as follows and as shown in table II.

TABLE II. Assignment of military part numbers and general information.

Part no.	Use with symbol no.	Description
M24231/29-001	713 and 713.5	3-conductor straight or 90 degrees angled
M24231/29-002	713.6 and 713.8	4-conductor straight or 90 degrees angled
M24231/29-003	713.9	3-conductor (12-pin)
M24231/29-004	713.1	7-conductor
M24231/29-005	713.7	9-conductor
M24231/29-006	713.2	14-conductor straight or 90 degrees angled
M24231/29-007	713.3	24-conductor straight or 90 degrees angled
M24231/29-008	713.4	30-conductor straight or 90 degrees angled
M24231/29-009	713.10	40-conductor straight or 90 degrees angled
M24231/29-010	None	5-conductor straight or 90 degrees angled

(a) Contact surfaces of socket contacts, items 11 and 12 shall be free from dirt, grease or any other contaminants at all times before, during, and after molding.

(b) Sleeving shall be installed over each socket with end of sleeve 0.031 inch from face of plug. Fit shall be snug and precautions shall be taken to prevent entrance of polyurethane into socket contact and spring clip area.

(c) Solder cups shall be aligned to facilitate soldering. Surfaces of all socket contacts shall be free of any molding compound or any other contaminant. Removal of contaminants after molding shall not be allowed as cleaning may damage or remove gold plating.

(d) Insert assemblies, items 01 through 09 and 28, shall be molded as follows:

(1) Slip plastic sleeving over socket contacts.
(2) Assemble socket contacts on loading pins in pin aligning head.
(3) Push the plastic sleeving down until sleeve rests on shoulder of loading pin.
(4) Place preform ring on pin aligning head, as shown on figure 1, and fill the area around socket contacts with molding compound, item 10, to top of preform ring.
(5) Allow to stand until all entrapped air rises to the surface, approximately 15 minutes. Refill to top of preform ring, if necessary. Keep solder cups clean.

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REQUIREMENTS: (Continued)

(6) Place assembly into 180°F oven and cure for 3 hours.
(7) After curing, allow assemblies to cool until preform molded insert can be removed from preform ring and pin aligning head.

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TABLE III. Parts list.

Item 1/ no.	Description 2/	Quantity per assembly M24231/29- 3/										Material/specification	Remarks
		001	002	003	004	005	006	007	008	009	010		
01	Insert assembly	-										See table I	Figures 1 and 2
02	Insert assembly		-									See table I	Figures 1 and 3
03	Insert assembly			-								See table I	Figures 1 and 4
04	Insert assembly				-							See table I	Figures 1 and 6
05	Insert assembly					-						See table I	Figures 1 and 7
06	Insert assembly						-					See table I	Figures 1 and 8
07	Insert assembly							-				See table I	Figures 1 and 9
08	Insert assembly								-			See table I	Figures 1 and 10
09	Insert assembly									-		See table I	Figures 1 and 11
10	Compound, molding	As required										Polyurethane, MIL-M-24041 1/	Figure 1
11	Contact, socket, size 16	3	4		7	9	14	24	30	40	5	2/	Figures 1 and 2
12	Contact, socket, size 12			3								2/	Figures 1 and 4
13	Sleeving, size 9	3	4		7	9	14	24	30	40	5	Plastic, MIL-I-631 3/	Figures 1 and 2
14	Sleeving, size 6			3								Plastic, MIL-I-631 3/	Figures 1 and 4
15	Insert assembly										-	See table I	Figures 1 and 5

See footnotes at top of next page.

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1/ The polyurethane compound used in molding the preform molding inserts shall be in accordance with category I of MIL-M-24041, clear or amber.

2/ Contact socket, item 11, shall be Bendix Aviation Corp. size 16M, short, part no. 10-101515-15S, gold plated, L. L. Rowe Company, part no. 40097, connector technology part no. SCA1315-1 or equal. Contact socket, item 12, shall be Bendix, size 12 solder type, part no. 10-33644, silver plated, followed by gold plate, not less than 0.0001 inch thick, or equal. Any contact socket submitted as equal shall be subject to NAVSEA approval.

3/ Plastic sleeving, items 13 and 14, shall be in accordance with type F, form Ua, grade A, class II, category I, color transparent (clear) of MIL-I-631.

NOTES:

1. All dimensions are in inches.
2. Unless otherwise indicated, tolerances shall be as follows:

   (a) Decimals - plus or minus 0.005 inch.
   (b) Fractions - plus or minus 1/64 inch.
   (c) Angular dimensions - plus or minus 1/2 degree.

Preparing activity:
Navy - SH
(Project 5935-N279-29)

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